SG No. 182 D_6^6 $P6_322$ [hexagonal]

* plus set: $+[0,\ 0,\ 0]$

* Wyckoff site: `2a`, site symmetry: `32.`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `32.`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ 0]$	`[1,2,3,-7,-8,-9]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,5,6,-10,-11,-12]`

Table 2: Wyckoff bond: `6b@2a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$[0,\ 0,\ 0]$	`[1,-8]`
2	$[-2X,\ -X,\ Z]$	$[0,\ 0,\ 0]$	`[2,-7]`
3	$[X,\ -X,\ Z]$	$[0,\ 0,\ 0]$	`[3,-9]`
4	$[-X,\ -2X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,-11]`
5	$[2X,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5,-10]`
6	$[-X,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[6,-12]`

Table 3: Wyckoff bond: `6c@2a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$[0,\ 0,\ 0]$	`[1,8]`
2	$[0,\ X,\ 0]$	$[0,\ 0,\ 0]$	`[2,7]`
3	$[-X,\ -X,\ 0]$	$[0,\ 0,\ 0]$	`[3,9]`
4	$[-X,\ 0,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,11]`
5	$[0,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5,10]`
6	$[X,\ X,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[6,12]`

Table 4: Wyckoff bond: `12d@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ 0]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ 0]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$[0,\ 0,\ 0]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[0,\ 0,\ 0]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$[0,\ 0,\ 0]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[12]`

* Wyckoff site: `2b`, site symmetry: `3.2`

Table 5: Wyckoff bond: `2a@2b`, bond symmetry: `3.2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,5,6,-7,-8,-9]`

Table 6: Wyckoff bond: `6b@2b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[6,-9]`

Table 7: Wyckoff bond: `6c@2b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,11]`
2	$[-2X,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[2,10]`
3	$[X,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3,12]`
4	$[-X,\ -2X,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,8]`
5	$[2X,\ X,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,7]`
6	$[-X,\ X,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[6,9]`

Table 8: Wyckoff bond: `12d@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[12]`

* Wyckoff site: `2c`, site symmetry: `3.2`

Table 9: Wyckoff bond: `2a@2c`, bond symmetry: `3.2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[4,5,6,-7,-8,-9]`

Table 10: Wyckoff bond: `6b@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[6,-9]`

Table 11: Wyckoff bond: `6c@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,11]`
2	$[-2X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[2,10]`
3	$[X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[3,12]`
4	$[-X,\ -2X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[4,8]`
5	$[2X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[5,7]`
6	$[-X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[6,9]`

Table 12: Wyckoff bond: `12d@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[12]`

* Wyckoff site: `2d`, site symmetry: `3.2`

Table 13: Wyckoff bond: `2a@2d`, bond symmetry: `3.2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[4,5,6,-7,-8,-9]`

Table 14: Wyckoff bond: `6b@2d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[6,-9]`

Table 15: Wyckoff bond: `6c@2d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[1,11]`
2	$[-2X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[2,10]`
3	$[X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[3,12]`
4	$[-X,\ -2X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[4,8]`
5	$[2X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[5,7]`
6	$[-X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[6,9]`

Table 16: Wyckoff bond: `12d@2d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[12]`

* Wyckoff site: `4e`, site symmetry: `3..`

Table 17: Wyckoff bond: `4a@4e`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,2,3]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4,5,6]`
3	$[0,\ 0,\ -Z]$	$[0,\ 0,\ -z]$	`[7,8,9]`
4	$[0,\ 0,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[10,11,12]`

Table 18: Wyckoff bond: `12b@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ z]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ z]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$[0,\ 0,\ -z]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[0,\ 0,\ -z]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$[0,\ 0,\ -z]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[12]`

* Wyckoff site: `4f`, site symmetry: `3..`

Table 19: Wyckoff bond: `4a@4f`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[4,5,6]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[10,11,12]`

Table 20: Wyckoff bond: `12b@4f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[10]`

continued ...

Table 20

No.	vector	center	mapping
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[12]`

* Wyckoff site: `6g`, site symmetry: `.2.`

Table 21: Wyckoff bond: `6a@6g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$[x,\ 0,\ 0]$	`[1,-8]`
2	$[-2X,\ -X,\ Z]$	$[0,\ x,\ 0]$	`[2,-7]`
3	$[X,\ -X,\ Z]$	$[-x,\ -x,\ 0]$	`[3,-9]`
4	$[-X,\ -2X,\ Z]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,-11]`
5	$[2X,\ X,\ Z]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[5,-10]`
6	$[-X,\ X,\ Z]$	$\left[x,\ x,\ \frac{1}{2}\right]$	`[6,-12]`

Table 22: Wyckoff bond: `6b@6g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$[x,\ 0,\ 0]$	`[1,8]`
2	$[0,\ X,\ 0]$	$[0,\ x,\ 0]$	`[2,7]`
3	$[-X,\ -X,\ 0]$	$[-x,\ -x,\ 0]$	`[3,9]`
4	$[-X,\ 0,\ 0]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,11]`
5	$[0,\ -X,\ 0]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[5,10]`
6	$[X,\ X,\ 0]$	$\left[x,\ x,\ \frac{1}{2}\right]$	`[6,12]`

Table 23: Wyckoff bond: `12c@6g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ 0,\ 0]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ x,\ 0]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[-x,\ -x,\ 0]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[x,\ x,\ \frac{1}{2}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$[0,\ x,\ 0]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$[x,\ 0,\ 0]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$[-x,\ -x,\ 0]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[x,\ x,\ \frac{1}{2}\right]$	`[12]`

* Wyckoff site: `6h`, site symmetry: `..2`

Table 24: Wyckoff bond: `6a@6h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[6,-9]`

Table 25: Wyckoff bond: `6b@6h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,11]`
2	$[-2X,\ -X,\ 0]$	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[2,10]`
3	$[X,\ -X,\ 0]$	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[3,12]`
4	$[-X,\ -2X,\ 0]$	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[4,8]`
5	$[2X,\ X,\ 0]$	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[5,7]`
6	$[-X,\ X,\ 0]$	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[6,9]`

Table 26: Wyckoff bond: `12c@6h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[12]`

* Wyckoff site: `12i`, site symmetry: `1`

Table 27: Wyckoff bond: `12a@12i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[-y,\ x-y,\ z]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[-x+y,\ -x,\ z]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	`[6]`

continued ...

Table 27

No.	vector	center	mapping
7	$[Y,\, X,\, -Z]$	$[y,\, x,\, -z]$	[7]
8	$[X-Y,\, -Y,\, -Z]$	$[x-y,\, -y,\, -z]$	[8]
9	$[-X,\, -X+Y,\, -Z]$	$[-x,\, -x+y,\, -z]$	[9]
10	$[-Y,\, -X,\, -Z]$	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[10]
11	$[-X+Y,\, Y,\, -Z]$	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	[11]
12	$[X,\, X-Y,\, -Z]$	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	[12]